

Practical 2

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
lahari@DESKTOP-AIC33F7: ~/ x + v
lahari@DESKTOP-AIC33F7:~$ mkdir practical2
mkdir: cannot create directory 'practical2': File exists
lahari@DESKTOP-AIC33F7:~$ cd practical2
lahari@DESKTOP-AIC33F7:~/practical2$ nano practical2.c
lahari@DESKTOP-AIC33F7:~/practical2$ gcc practical2.c -o practical2
lahari@DESKTOP-AIC33F7:~/practical2$ ./practical2
Enter source file name: ^C
lahari@DESKTOP-AIC33F7:~/practical2$ echo "Hello World" > sample.txt
lahari@DESKTOP-AIC33F7:~/practical2$ ./practical2
Enter source file name: sample.txt
Enter destination file name: copy.txt

File copied successfully.
lahari@DESKTOP-AIC33F7:~/practical2$ cat copy.txt
Hello World
lahari@DESKTOP-AIC33F7:~/practical2$ |
```